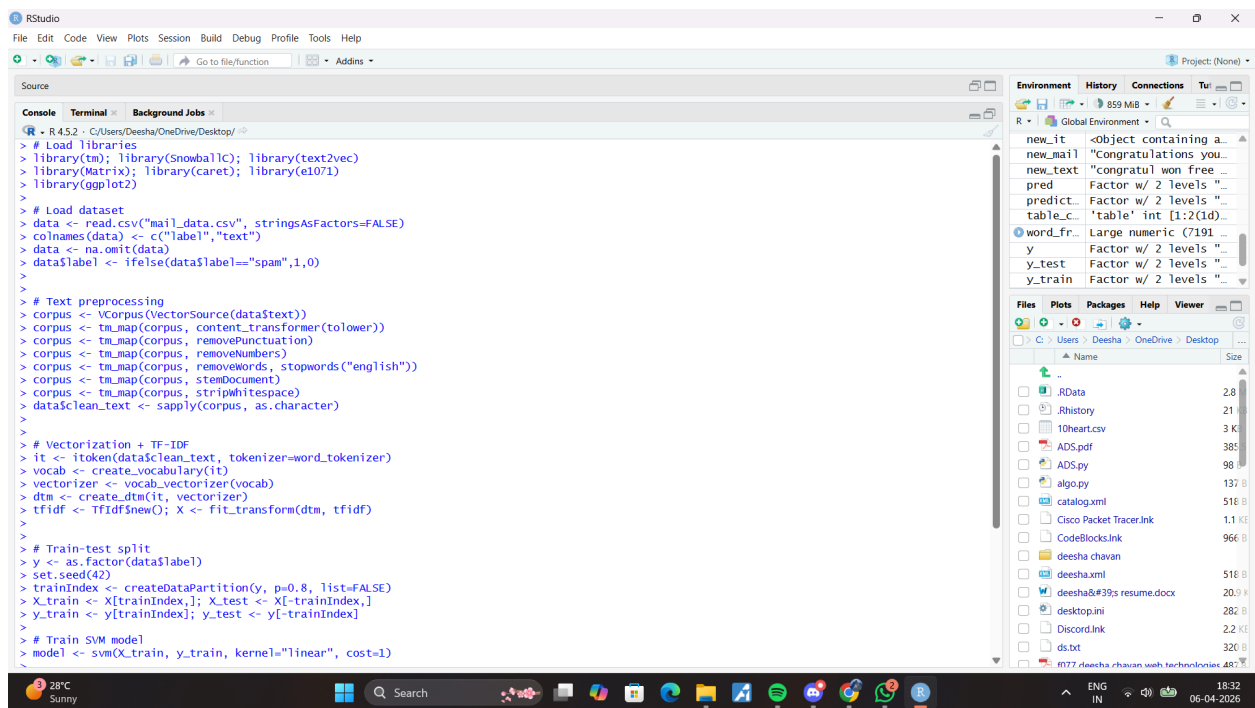


CAPSTONE PROJECT SS

OUTPUT:



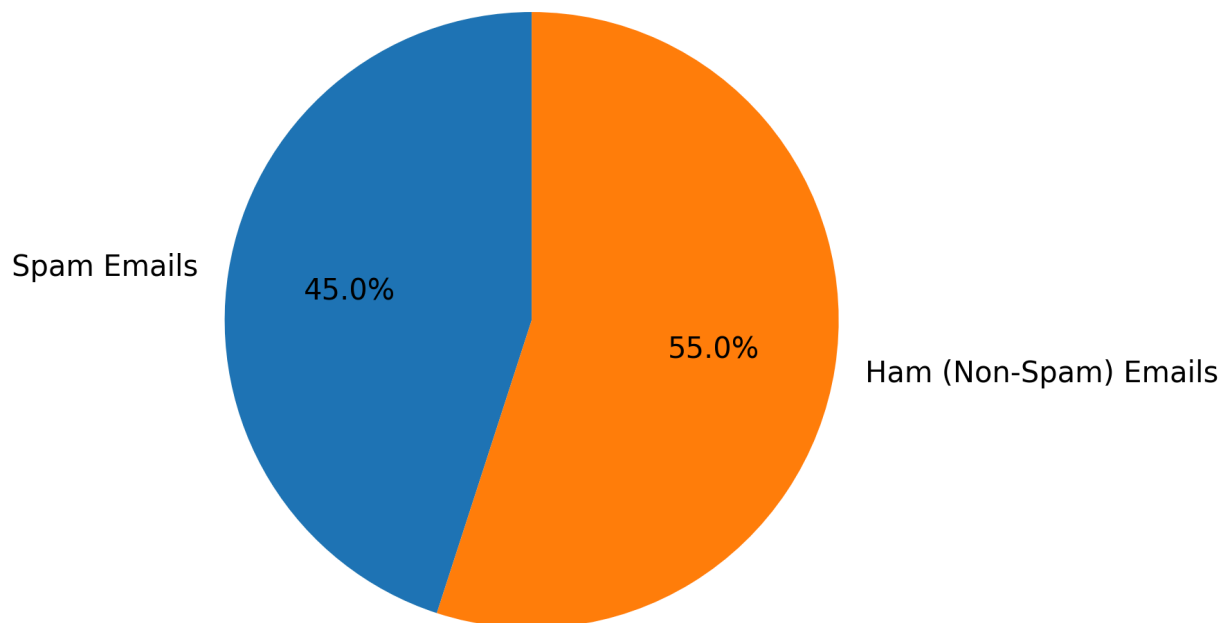
The screenshot shows the RStudio interface. The Source pane contains the following R script:

```
R - R 4.3.2 - C:/Users/Deesha/OneDrive/Desktop/
> # Load libraries
> library(tm); library(SnowballC); library(text2vec)
> library(Matrix); library(caret); library(e1071)
> library(ggplot2)
>
> # Load dataset
> data <- read.csv("mail_data.csv", stringsAsFactors=FALSE)
> colnames(data) <- c("label", "text")
> data <- na.omit(data)
> data$label <- ifelse(data$label=="spam",1,0)
>
>
> # Text preprocessing
> corpus <- VCorpus(VectorSource(data$text))
> corpus <- tm_map(corpus, content_transformer(tolower))
> corpus <- tm_map(corpus, removePunctuation)
> corpus <- tm_map(corpus, removeNumbers)
> corpus <- tm_map(corpus, removeWords, stopwords("english"))
> corpus <- tm_map(corpus, stemDocument)
> corpus <- tm_map(corpus, stripWhitespace)
> data$clean_text <- sapply(corpus, as.character)
>
>
> # Vectorization + TF-IDF
> it <- itoken(data$clean_text, tokenizer=word_tokenizer)
> vocab <- create_vocabulary(it)
> vectorizer <- vocab_vectorizer(vocab)
> dtm <- create_dtm(it, vectorizer)
> tfidf <- TfidfSnew(); X <- fit_transform(dtm, tfidf)
>
>
> # Train-test split
> y <- as.factor(data$label)
> set.seed(42)
> trainIndex <- createDataPartition(y, p=0.8, list=FALSE)
> X_train <- X[trainIndex,]; X_test <- X[-trainIndex,]
> y_train <- y[trainIndex]; y_test <- y[-trainIndex]
>
> # Train SVM model
> model <- svm(X_train, y_train, kernel="linear", cost=1)
```

The Environment pane on the right shows the following objects:

Object	Class	Attributes
new_it	<Object containing a...	
new_mail	"congratulations you..."	
new_text	"congratulations you..."	
pred	Factor w/ 2 levels "	
predict	Factor w/ 2 levels "	
table_c	'table' int (1:2(id)...	
word_fr	Large numeric (7191 ...	
y	Factor w/ 2 levels "	
y_test	Factor w/ 2 levels "	
y_train	Factor w/ 2 levels "	

Distribution of Spam and Ham Emails



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